THINK UNLIMITED PRESENTS

CLASS 10 MATHS
MADE EASY

CBSE Board Exam Edition 2026/27

80/80
TARGET SCORE

"Maths se darr? Ab nahi."

Formula • Trick • Example — Sab ek hi jagah

WHAT IS INSIDE THIS 20-PAGE BLUEPRINT:

- All 14 NCERT Chapters mapped to CBSE 2026/27 syllabus
- 100% Formulas with "When-to-Apply" conditions
- Shortcut Tricks & Vedic cross-checks for MCQs
- "Marks Bachao" alerts: Top mistakes students make
- 3-Hour Exam Day Time Management & Step-marking guide

Chalo, maths mein full marks laayein!

Student Edition • Support WhatsApp: +91 9243791529 • Email: a2043728022@gmail.com

CBSE Class 10 Maths: Blueprint & Marks Weightage (80 Marks)

Board exams mein smart study sabse zaroori hai. Pata karo kahan se kitne marks aate hain:

Unit I: Number Systems (Real Numbers)	**6 Marks**	High ROI (Quick 6)
Unit II: Algebra (Polynomials, Linear Eq, Quadratic Eq, AP)	**20 Marks**	Core Foundation
Unit III: Coordinate Geometry	**6 Marks**	Guaranteed Full Marks
Unit IV: Geometry (Triangles, Circles)	**15 Marks**	Theorem Practice
Unit V: Trigonometry (Basics, Identities, Heights & Distances)	**12 Marks**	High Scoring Formulas
Unit VI: Mensuration (Areas Related to Circles, Surface Areas)	**10 Marks**	Direct Formula Calculation
Unit VII: Statistics & Probability	**11 Marks**	Easiest 11 Marks

THE 80/80 TOPPER STRATEGY:

1. Start with Statistics + Probability + Real Numbers (Total = 23 Marks). Yeh 2 din mein khatam hote hain.
2. Coordinate Geometry (6 Marks) formula based hai - distance, section formula se pure marks aate hain.
3. Algebra (20 Marks) mein quadratic equations aur AP ke direct shortcuts use karo.
4. Geometry theorems (BPT Theorem & Tangent Theorem) ko 5-5 baar likh kar yaad karo.

& þ MARKS BACHAO ALERT (Common Board Mistake):

Student mistake: Direct calculation mein rush karna bina formula likhe. CBSE step-marking deta hai! Formula likhne ka alag se 0.5 to 1 mark milta hai chahe final answer mein calculation slip ho.

Chapter 1: Real Numbers

Expected Marks: 6 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• Numbers are either rational (p/q, terminating or repeating decimal) or irrational (non-terminating, non-repeating).

Ø=ÜÐ Must-Know Board Formulas:

• Fundamental Theorem of Arithmetic: Every composite number = unique product of primes.

• HCF(a, b) × LCM(a, b) = a × b (Only valid for TWO numbers, not three!).

• If prime p divides a^2, then p divides a.

&¡ Topper Shortcut Trick:

• To find HCF of two numbers quickly, use prime factorization with the lowest powers of common prime factors. For LCM, take highest powers of all prime factors.

' þ Solved Board Model Question:

• Q: If HCF(306, 657) = 9, find LCM(306, 657).

Ans: LCM = (306 × 657) / 9 = 34 × 657 = 22,338.

& þ MARKS BACHAO ALERT (Common Board Mistake):

Applying HCF(a,b,c) × LCM(a,b,c) = a × b × c for THREE numbers! Yeh rule sirf 2 numbers par kaam karta hai.

Ø<ß¯ Quick Check Question (Khud Try Karo):

• Find HCF and LCM of 12 and 18, and verify that HCF × LCM = 12 × 18.

Chapter 2: Polynomials

Expected Marks: 4-5 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• A polynomial of degree n has at most n real zeroes (where the graph intersects the x-axis).

Ø=ÜÐ Must-Know Board Formulas:

• Quadratic Polynomial: p(x) = ax² + bx + c

• Sum of zeroes (; ² ²) = -b/a = -(coefficient of x) / (coefficient of x²)

• Product of zeroes (; ²) = c/a = (constant term) / (coefficient of x²)

• Forming quadratic polynomial: k [x² - (; ² ²)x + ; ²]

&¡ Topper Shortcut Trick:

• Memory Trick: "Sum minus, product plain". Zeroes ka sum hamesha minus sign se shuru hota hai (-b/a).

' þ Solved Board Model Question:

• Q: Find a quadratic polynomial whose sum and product of zeroes are -3 and 2.
Ans: p(x) = x² - (-3)x + 2 = x² + 3x + 2.

& þ MARKS BACHAO ALERT (Common Board Mistake):

Formula mein minus sign bhool jana! x² - (; ³²)x + ; ² hota hai, x² + (; ³²)x nahi.

Ø<ß¯ Quick Check Question (Khud Try Karo):

• If ; Â ² are zeroes of x² - 5x + 6, find the value of ; " ² ²² using (; ³²)² - 2; ².

Chapter 3: Pair of Linear Equations in Two Variables

Expected Marks: 5-6 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• Two straight lines a •x + b •y + c • = 0 and a ,x + b ,y + c , = 0 can intersect at 1 point, be parallel, or coincide.

Ø=ÜÐ Must-Know Board Formulas:

• Unique solution (Intersecting): a •/a , "` b •/b , [Consistent]

• Infinitely many solutions (Coincident): a •/a , = b •/b , = c •/c , [Consistent & Dependent]

• No solution (Parallel lines): a •/a , = b •/b , "` c •/c , [Inconsistent]

&¡ Topper Shortcut Trick:

• Elimination Method shortcut: Multiply equations to equate the smaller coefficient first to minimize arithmetic errors.

' þ Solved Board Model Question:

• Q: For what value of k will 2x + 3y = 7 and (k-1)x + (k+2)y = 3k have infinite solutions?

Ans: 2/(k-1) = 3/(k+2) = 7/3k. Cross-multiply: 2(k+2) = 3(k-1) => 2k+4 = 3k-3 => k = 7.

& þ MARKS BACHAO ALERT (Common Board Mistake):

Standard form check karna bhoolna! Always ensure both equations have constant term on the same side (either both RHS or both LHS) before taking c •/c ,.

Ø<ß¯ Quick Check Question (Khud Try Karo):

• Solve for x and y: 2x + 3y = 11 and 2x - 4y = -24.

Chapter 4: Quadratic Equations

Expected Marks: 6 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• Equations of the form $ax^2 + bx + c = 0$ (a "` 0). The discriminant D decides the nature of the roots.

Ø=ÜÐ Must-Know Board Formulas:

• Discriminant: $D = b^2 - 4ac$

• Quadratic Formula (Sridharacharya): $x = (-b \pm \sqrt{D}) / (2a)$

• If D > 0: Two distinct real roots

• If D = 0: Two equal real roots ($x = -b / 2a$)

• If D < 0: No real roots

&¡ Topper Shortcut Trick:

• Topper Shortcut: If a + b + c = 0, one root is ALWAYS 1 and the other root is c/a. Example: $3x^2 - 5x + 2 = 0$ (3 - 5 + 2 = 0), so roots are 1 and 2/3 directly!

' þ Solved Board Model Question:

• Q: Find roots of $2x^2 - 7x + 3 = 0$ using quadratic formula.
Ans: a=2, b=-7, c=3. $D = (-7)^2 - 4(2)(3) = 49 - 24 = 25$. $x = (7 \pm 5)/4$ => x = 3 or x = 1/2.

& þ MARKS BACHAO ALERT (Common Board Mistake):

b^2 calculate karte waqt $(-7)^2$ ko -49 likhna! Negative number ka square hamesha POSITIVE hota hai (+49).

Ø<ß¯ Quick Check Question (Khud Try Karo):

• If $2x^2 + kx + 3 = 0$ has equal roots, find the value of k.

Chapter 5: Arithmetic Progressions (AP)

Expected Marks: 5-6 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• A sequence where each term differs from the previous by a constant common difference d.

Ø=ÜÐ Must-Know Board Formulas:

• Common difference: d = a , - a • = a ™ - a ™ ‹ •

• n-th term: a ™ = a + (n - 1)d

• Sum of first n terms: S ™ = n/2 [2a + (n - 1)d] = n/2 [a + l] (where l = last term)

• n-th term from the end: l - (n - 1)d

• Three consecutive terms in AP: (a - d), a, (a + d)

&¡ Topper Shortcut Trick:

• Finding n-th term from S ™: a ™ = S ™ - S ™ ‹ •. Also, S • = a •.

' þ Solved Board Model Question:

• Q: How many two-digit numbers are divisible by 3?

Ans: AP: 12, 15, ..., 99. a=12, d=3, a ™=99. 99 = 12 + (n-1)3 => 87 = (n-1)3 => n-1 = 29 => n = 30.

& þ MARKS BACHAO ALERT (Common Board Mistake):

Confusing "n" with "a ™". n is the number of terms (must be a positive integer), whereas a ™ is the value of the term.

Ø<ß¯ Quick Check Question (Khud Try Karo):

• Find the 20th term from the last term of the AP: 3, 8, 13, ..., 253.

Chapter 6: Triangles

Expected Marks: 8-9 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• Two figures are similar if their corresponding angles are equal and corresponding sides are in the same ratio.

Ø=ÜÐ Must-Know Board Formulas:

• Basic Proportionality Theorem (Thales Theorem): If DE || BC, then AD/DB = AE/EC.

• Converse of BPT: If AD/DB = AE/EC, then DE || BC.

• Similarity Criteria: AAA (or AA), SSS, SAS.

&¡ Topper Shortcut Trick:

• Always write similarity with correct vertex correspondence! If 9D $2 â ”DEF, then AB/DE = BC/EF = AC/DF.

' þ Solved Board Model Question:

• Q: In 9D $0, DE || BC. If AD = 1.5 cm, DB = 3 cm, AE = 1 cm, find EC.

Ans: By BPT, AD/DB = AE/EC => 1.5/3 = 1/EC => 1/2 = 1/EC => EC = 2 cm.

& þ MARKS BACHAO ALERT (Common Board Mistake):

Board exam mein BPT Theorem ka statement aur "Given, To Prove, Construction, Proof" format chodna. Har heading ke marks alag hote hain!

Ø<ß¯ Quick Check Question (Khud Try Karo):

• State and prove Basic Proportionality Theorem (Thales Theorem).

Chapter 7: Coordinate Geometry

Expected Marks: 6 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• Locating points and shapes on the Cartesian plane using coordinates (x, y).

Ø=ÜÐ Must-Know Board Formulas:

• Distance Formula: d = " [(x , - x •)² + (y , - y •)²]

• Distance of point (x, y) from origin: " (x² + y²)

• Section Formula (Internal division m •:m ,):
x = (m •x , + m ,x •) / (m • + m ,), y = (m •y , + m ,y •) / (m • + m ,)

• Mid-point Formula: ((x • + x ,)/2, (y • + y ,)/2)

&¡ Topper Shortcut Trick:

• Ratio Trick: Whenever ratio is to be found, always assume the ratio to be k : 1 instead of m • : m ,. Calculations become 2x faster.

' þ Solved Board Model Question:

• Q: Find coordinates of point dividing (4, -3) and (8, 5) in ratio 3:1.
Ans: x = (3(8) + 1(4))/(3+1) = 28/4 = 7; y = (3(5) + 1(-3))/(3+1) = 12/4 = 3. Point is (7, 3).

& þ MARKS BACHAO ALERT (Common Board Mistake):

Section formula mein coordinates cross multiply na karna! m • goes with x ,, and m , goes with x •.

Ø<ß¯ Quick Check Question (Khud Try Karo):

• Find the ratio in which the y-axis divides the line segment joining (5, -6) and (-1, -4).

Chapter 8: Introduction to Trigonometry

Expected Marks: 6 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• Trigonometric ratios are defined for acute angles in a right-angled triangle.

Ø=ÜÐ Must-Know Board Formulas:

• sin ;, Ò ô,Â 6÷2 ¸ = B/H, tan ;, Ò ô

• cosec ;, Ò ÷6–â ¸, sec ;, Ò ö6÷2 ¸, cot ;, Ò ÷F â ¸

• Standard Table:
sin: 0°=0, 30°=1/2, 45°=1/" 2, 60°=" 3/2, 90°=1
cos: 0°=1, 30°=" 3/2, 45°=1/" 2, 60°=1/2, 90°=0
tan: 0°=0, 30°=1/" 3, 45°=1, 60°=" 3, 90°=Not Defined

&¡ Topper Shortcut Trick:

• Hand / Finger rule for sin values: " (fingers below) / 2. Thumb = 0°, Index = 30°, Middle = 45°, Ring = 60°, Pinky = 90°!

' þ Solved Board Model Question:

• Q: Evaluate: 2 tan² 45° + cos² 30° - sin² 60°.
Ans: 2(1)² + (" 3/2)² - (" 3/2)² = 2 + 3/4 - 3/4 = 2.

& þ MARKS BACHAO ALERT (Common Board Mistake):

Assuming sin(A + B) = sin A + sin B. This is NEVER true! Trigonometric ratio is not a multiplication of sin with angle.

Ø<ß¯ Quick Check Question (Khud Try Karo):

• If tan A = 4/3, find all other trigonometric ratios of angle A.

Chapter 9: Trigonometric Identities

Expected Marks: 4-5 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• Identities that hold true for all values of angles in a right triangle.

Ø=ÜÐ Must-Know Board Formulas:

• Identity 1: sin² ;, ² 6÷;" ¸ = 1 => sin² ;, Ò Ò 6÷;" ¸, cos² ;, Ò Ò 6–ë" ¸

• Identity 2: 1 + tan² ;, Ò 6V;" ¸ => sec² ;, Ò F ë" ¸ = 1

• Identity 3: 1 + cot² ;, Ò 6÷6V;" ¸ => cosec² ;, Ò 6÷K" ¸ = 1

• tan ;, Ò 6–â ¸ / cos ;,Â 6÷B ¸ = cos ;, ò 6–â ¸

&¡ Topper Shortcut Trick:

• Golden Rule of Trigonometric Proofs: When stuck, convert every term (tan, cot, sec, cosec) into sin ;, æB 6÷2 ¸. 95% of questions resolve in 3 steps!

' þ Solved Board Model Question:

• Q: Prove that (sin ;, ² 6÷6V2 ¸)² + (cos ;, ² 6V2 ¸)² = 7 + tan² ;, ² 6÷K" ¸.

Ans: Expand LHS: sin²;, ² " ² 6÷6V;#¸ + cos²;, ² " ² 6V;#¸ = (sin²;,¶6÷;#¸) + 4 + (1+cot²;,' ² ƒ ·F ë#¸) = 1 + 4 + 1 + cot²;, ² ² tan²;, Ò r ² F ë#¸ + cot²;,à

& þ MARKS BACHAO ALERT (Common Board Mistake):

Dividing by zero or cancelling terms without justifying. Also forgetting sec²;, Ò F ë#¸ = (sec;, Ò F ã¸)(sec;, ² F ã¸) = 1.

Ø<ß¯ Quick Check Question (Khud Try Karo):

• Prove that: (1 + tan² A) / (1 + cot² A) = tan² A.

Chapter 10: Some Applications of Trigonometry

Expected Marks: 4 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• Heights and distances calculated using line of sight, angle of elevation, and angle of depression.

Ø=ÜÐ Must-Know Board Formulas:

• Angle of elevation: Looking UP from horizontal.

• Angle of depression: Looking DOWN from horizontal. (Angle of depression = Angle of elevation from base by alternate interior angles).

• tan ;, Ò †V–v‡B ò F—7F æ6R †Ö÷7B 6öÖÖöâ atio in word problems).

&¡ Topper Shortcut Trick:

• Values to memorize: " 3 "H 1.732, " 2 "H 1.414. Always use given values in question if specified.

' þ Solved Board Model Question:

• Q: A tower stands vertically on ground. From a point 15 m away from foot, angle of elevation is 60°. Find height.
Ans: tan 60° = h / 15 => " 3 = h / 15 => h = 15" 3 m "H 25.98 m.

& þ MARKS BACHAO ALERT (Common Board Mistake):

Diagram mein observer ki eye-level aur ground horizontal mein confusion. Diagram carries 1 full mark in Board marking scheme!

Ø<ß¯ Quick Check Question (Khud Try Karo):

• A ladder 15 m long reaches a window 12 m above ground. Find the distance of foot of ladder from wall.

Chapter 11: Circles

Expected Marks: 6 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• A tangent to a circle is a line that intersects the circle at only one point.

Ø=ÜÐ Must-Know Board Formulas:

• Theorem 10.1: The tangent at any point of a circle is perpendicular to the radius through the point of contact (OP "¥ AB).

• Theorem 10.2: The lengths of tangents drawn from an external point to a circle are equal (PQ = PR).

• The center lies on the bisector of the angle between the two tangents.

&¡ Topper Shortcut Trick:

• External Tangent Quadrangle: Opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the center (Sum = 180°).

' þ Solved Board Model Question:

• Q: If tangents PA and PB from point P to a circle with center O are inclined to each other at 80°, find " POA.
Ans: " AOB = 180° - 80° = 100°. In right triangles, OP bisects " AOB, so " POA = 100° / 2 = 50°.

& þ MARKS BACHAO ALERT (Common Board Mistake):

Forgetting that radius meets tangent at exactly 90°. Always mark 90° in the diagram immediately.

Ø<ß¯ Quick Check Question (Khud Try Karo):

• Prove that the tangents drawn at the ends of a diameter of a circle are parallel.

Chapter 12: Areas Related to Circles

Expected Marks: 4-5 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• Calculating perimeter and areas of circles, sectors, and segments.

Ø=ÜÐ Must-Know Board Formulas:

• Circumference = 2< , Area = < +

• Length of arc of sector with angle ;ƒ l = (;, ò 3c ’ r #Àr

• Area of sector with angle ;ƒ (;, ò 3c ’ r Àr²

• Area of minor segment = Area of sector - Area of corresponding triangle

• Area of triangle with angle ;, &WGpeen two radii: 1/2 × r² × sin ;€

&¡ Topper Shortcut Trick:

• Area of triangle for segment: Use Area = 1/2 × r² × sin ;, F—&V7FÇ’ –b ¸ = 60°, sin 60° = " 3/2. If ;, Ò “ Â 6–â “ Ò à

' þ Solved Board Model Question:

• Q: Find area of sector of circle with radius 6 cm if angle is 60°.

Ans: Area = (60/360) × (22/7) × 6 × 6 = 1/6 × 22/7 × 36 = 132/7 cm² "H 18.86 cm².

& þ MARKS BACHAO ALERT (Common Board Mistake):

Confusing sector (pizza slice) with segment (crust cut by chord). Read the question carefully!

Ø<ß¯ Quick Check Question (Khud Try Karo):

• Find the area of a quadrant of a circle whose circumference is 22 cm.

Chapter 13: Surface Areas and Volumes

Expected Marks: 5-6 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

- Combination of solids: Cube, Cuboid, Cylinder, Cone, Sphere, Hemisphere.

Ø=ÜÐ Must-Know Board Formulas:

- Cylinder: CSA = 2< &,Â TSA = 2< "‡" ² ,'Â Volume = < +&€
- Cone: Slant height l = " (r² + h²), CSA = < l, TSA = < "†Â ² "'Â Volume = 1/3 < +&€
- Sphere: Surface Area = 4< +"Â Volume = 4/3 < +0
- Hemisphere: CSA = 2< +"Â TSA = 3< +"Â Volume = 2/3 < +0

&¡ Topper Shortcut Trick:

- When one solid is melted and recast into another, VOLUME remains constant! Equate Volume • = Volume ,.

' þ Solved Board Model Question:

- Q: 2 cubes each of volume 64 cm³ are joined end to end. Find surface area of resulting cuboid.

Ans: Side of cube a = ³" 64 = 4 cm. Cuboid: l=8 cm, b=4 cm, h=4 cm. TSA = 2(lb+bh+hl) = 2(32+16+32) = 160 cm².

& þ MARKS BACHAO ALERT (Common Board Mistake):

When two solids are joined, their common face is HIDDEN! Do not add TSA of individual solids; add their visible CURVED surfaces (CSA • + CSA ,).

Ø<ß¯ Quick Check Question (Khud Try Karo):

- A metallic sphere of radius 4.2 cm is melted and recast into a cylinder of radius 6 cm. Find the height of the cylinder.

Chapter 14: Statistics & Probability

Expected Marks: 11 Marks

Ø=Ü¡ Simple Idea (Bina Ratta Samjho):

• Measures of central tendency (Mean, Median, Mode) and classical probability P(E) = favourable / total outcomes.

Ø=ÜÐ Must-Know Board Formulas:

• Mean (Direct): x0B Ò £(f bx b) / :6aÖ

• Mode = l + [(f • - f €) / (2f • - f € - f ,)] × h

• Median = l + [(N/2 - cf) / f] × h

• Empirical Relationship: 3 Median = Mode + 2 Mean

• Probability: P(E) = n(E) / n(S), where 0 "d P(E) "d 1 and P(E) + P(not E) = 1

&¡ Topper Shortcut Trick:

• Empirical Formula mnemonic: "3 Medians minus 2 Means equals Mode" (3M - 2M = M).

' þ Solved Board Model Question:

• Q: In a frequency distribution, Mean = 20 and Median = 22. Find Mode.

Ans: Mode = 3 Median - 2 Mean = 3(22) - 2(20) = 66 - 40 = 26.

& þ MARKS BACHAO ALERT (Common Board Mistake):

In Median calculation, choosing cf of the median class instead of cumulative frequency of the PRECEDING class!

Ø<ß¯ Quick Check Question (Khud Try Karo):

• A bag contains 3 red and 5 black balls. A ball is drawn at random. What is probability that the ball drawn is red?

The 3-Hour Board Exam Execution Plan

10:15 AM - 10:30 AM (15 Min Reading Time)
Do NOT pick up the pen! Scan all 38 questions. Mark easy questions with a pencil dot. Decide your choices in Section B, C, and D.

10:30 AM - 11:15 AM (Section A: 20 MCQs - 45 Min)
Spend 2 minutes per MCQ. Use shortcut tricks, elimination method, and option substitution. Do rough work clearly in the right margin.

11:15 AM - 11:45 AM (Section B: 5 Very Short - 30 Min)
2-mark questions. Write formula first, 2 steps of calculation, final answer in a box with correct units.

11:45 AM - 12:35 PM (Section C: 6 Short Answers - 50 Min)
3-mark questions. Draw neat pencil diagrams for geometry and trigonometry questions. Write reasons in brackets.

12:35 PM - 01:10 PM (Section D: 4 Long Answers - 35 Min)
5-mark questions (BPT Theorem, Surface Areas, Quadratic word problem). Step-marking is crucial.

01:10 PM - 01:25 PM (Section E: 3 Case Study Questions - 15 Min)
Read the case study paragraph fast; the questions are usually straightforward direct formulas.

01:25 PM - 01:30 PM (Final 5 Min Revision Check)
Double check question numbers, unit conversions (cm to m), and check if any sub-part was skipped.

& þ MARKS BACHAO ALERT (Common Board Mistake):
Never leave any question blank in Section D or Section E! Even writing the given values, drawing the diagram, and writing the formula fetches 1.5 to 2 marks out of 5.

Top 10 Silly Mistakes That Cost 10-15 Marks in Board Exam

1. Unit Mismatch: Adding cm and m without converting (e.g. radius in cm, height in m).
2. Missing Units in Final Answer: Forgetting cm^2, cm^3, or m/s in word problems (-0.5 mark deduction).
3. Negative Sign in Quadratic Formula: (-b ± " D) / 2a. If b = -4, then -b becomes -(-4) = +4.
4. Radius vs Diameter Trap: Reading diameter as 14 cm and taking r = 14 instead of r = 7 cm.
5. BPT Theorem Missing Justification: Writing AD/DB = AE/EC without writing "[By Thales Theorem DE || BC]".
6. Frequency vs Cumulative Frequency: Using cf instead of f in Mode or Median denominator.
7. Volume vs Surface Area: Adding TSA of cone and hemisphere instead of visible CSA.
8. Angle of Depression: Not showing the alternate angle equality in trigonometry diagram.
9. Probability out of range: Writing an answer greater than 1 or negative (Probability is strictly 0 to 1).
10. Rough Work Chaos: Doing rough calculations all over the page and copying wrong numbers into final answer.

THE 30-SECOND GOLDEN HABIT:

After finding the answer, reread the LAST sentence of the question. Did they ask for radius or diameter? Did they ask for speed of stream or boat in still water? Did they ask for age of son or father? This single habit saves 5 marks!

Quick Board Self-Test (Solve in 25 Minutes)

1. What is the HCF of smallest prime number and smallest composite number? [Ans: 2]

2. If ; æB ² are zeroes of 2x² - 4x + 5, what is ; ² ² + ; ²? [Ans: 2 + 5/2 = 9/2]

3. For what value of p do 2x + py = -5 and x + y = -1 have unique solution? [Ans: p "` 2]

4. Find the roots of x² - (" 3 + 1)x + " 3 = 0. [Ans: x = " 3, x = 1]

5. What is the 11th term of AP: -3, -1/2, 2, ...? [Ans: 22]

6. If 9D $2 â ”DEF and AB/DE = 2/3, find ratio of their perimeters. [Ans: 2:3]

7. Find distance between (a, b) and (-a, -b). [Ans: 2" (a² + b²)]

8. Evaluate: (sin² 30° + cos² 30°) / (sec² 45° - tan² 45°). [Ans: 1 / 1 = 1]

9. Tangents PA and PB are drawn to circle with radius 5 cm from P. If OP = 13 cm, find PA. [Ans: 12 cm]

10. If P(E) = 0.05, what is the probability of "not E"? [Ans: 0.95]

SCORE GUIDE & BENCHMARK:

- 9-10 Correct: Board 80/80 Topper Ready! You have strong conceptual clarity.
- 7-8 Correct: 70+ Marks Zone. Revise the shortcut tricks and alert boxes once more.
- Below 7: Re-read Chapter 1 to 5 formulas and solve 5 NCERT questions per chapter.

Final 24-Hour Board Formula Checklist

[ ] Real Numbers: HCF × LCM = a × b

[ ] Polynomials: ; ² ² = -b/a, ; ² = c/a

[ ] Linear Eq: Unique (a •/a , "` b •/b ,), Coincident, Parallel ratios

[ ] Quadratic Eq: D = b² - 4ac, x = (-b ± " D) / 2a

[ ] AP: a ™ = a + (n-1)d, S ™ = n/2[2a + (n-1)d] = n/2[a + l]

[ ] Triangles: BPT AD/DB = AE/EC & AAA, SAS similarity

[ ] Coordinate: Distance formula & Section formula m •:m ,

[ ] Trigonometry: sin, cos, tan values (0°, 30°, 45°, 60°, 90°)

[ ] Identities: sin²;, ² 6÷;#¸ = 1, 1 + tan²;, Ò 6V;#¸, 1 + cot²;, Ò 6÷6V;#¸

[ ] Circles: Radius "¥ Tangent, Equal tangents from external point

[ ] Areas: Sector = (;,ó3c “Àr², Segment = Sector - Triangle

[ ] Mensuration: Cylinder, Cone (l=" (r²+h²)), Sphere, Hemisphere

[ ] Statistics: Empirical formula: 3 Median = Mode + 2 Mean

[ ] Probability: P(E) = favourable / total, 0 "d P(E) "d 1

All The Best for Your Class 10 CBSE Board Exams! Ø<ß

Yaad rakhna: Confidence aati hai preparation se. Saare formulas ek paper par likho, 5-5 sawal lagao aur cool mind se paper do. You can easily score 80/80!

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